

Dynamics Test Plan

Team Name:	ROFASO
Project Name:	Damage Assessment Protocol (D.A.P.)
Sub-system Name:	Dynamics
Prepared By:	Geenadie Rathnayake, Kyle Wittcoff
Date:	5/7/2026
Test Dates:	5/17/2026
Document ID:	-

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1 Introduction

i Objective and Summary of Experiments

The objective of this test plan is to verify and validate the dynamics model of the Sealglider by comparing the simulated outputs to real nominal mission data. The dynamics code will take in parameters from the glider data files and then propagate the glider state over each time interval. These simulated outputs will be compared to the mission data either recorded by onboard sensors or the estimation outputted by the glider from its Hydrodynamic Model.

The outputs evaluated in this test are depth, vertical velocity, pitch, roll, heading, pitch rate, roll rate, and heading rate. These outputs are evaluated because they are used by the Diagonoser to assess glider behavior. The test will pass if each output is modeled within the allowable percent error specified by the following dynamics requirements.

- **DNR-1:** The modeled depth, $z(t)$, must be within ± 8.6 m of the onboard recorded depth at each evaluated time step during the dive.
- **DNR-2:** The modeled vertical velocity, $w(t)$, must be within ± 13 cm/s of the onboard recorded vertical velocity, dz/dt , at each evaluated time step during the dive.
- **DNR-3:** The modeled pitch angle must be within $\pm 53^\circ$ of the onboard recorded pitch angle at each evaluated time step during the dive.
- **DNR-4:** The modeled roll angle must be within $\pm 28^\circ$ of the onboard recorded roll angle at each evaluated time step during the dive.
- **DNR-5:** The modeled heading angle must be within $\pm 80^\circ$ of the onboard recorded heading angle at each evaluated time step during the dive.
- **DNR-6:** Pitch rate of a dive must be modeled within 3.3% error of onboard recorded pitch rate of a dive.
- **DNR-7:** Roll rate of a dive must be modeled within 3.3% error of onboard recorded roll rate of a dive.
- **DNR-8:** Heading rate of a dive must be modeled within 3.3% error of onboard recorded heading rate of a dive.

ii Related Procedures

List any test procedures associated with this plan.

Procedure ID	Description

2 Design of Experiments

i Parameters

The parameters depend on the dive, type of Seaglider used, and the fault case. Parameters that depend on the dive are $\psi_{desired}$, $\psi_{current}$, ρ , P , T , and T_0 . Parameters that depends on the glider properties are S , $\bar{c}$, b , V_{CB} , $V_{CG,bat}$, m_{bat} , V_{static} , $\mathbf{J}_s$, and m_s . For this test plan, SG 194 glider is used and it tests only nominal cases.

Variable	Description	Range	Source
S	Surface area of glider in XY plane	0.486 m ²	CAD
$\bar{c}$	Mean aerodynamic chord	0.173 m	CAD
b	Wing span	1 m	CAD
$\psi_{desired}$	Desired heading angle wrt magnetic north	varies with dive	log files
$\psi_{current}$	Heading angle of the glider wrt magnetic north	95° - 305°	log files
V_{CB}	Distance from the neutral axis to the Center of Buoyancy in the vertical (b_3) direction	-0.00362 m	SG194 WA COAST-230ct-2013 data sheet
$V_{CG,bat}$	Distance from the neutral axis to the CoM of the high voltage battery in the b_3 direction	-0.01082 m	SG 194 Mass and Inertia Matrix Calculation Document
m_{bat}	Battery mass	11.636 kg	ballasting data sheet
V_{static}	Static displaced volume of glider excluding bladder	m ³	
$\mathbf{M}_f$	Added mass matrix	$\begin{bmatrix} 3.310 & 0 & 0 \\ 0 & 72.005 & 0 \\ 0 & 0 & 72.005 \end{bmatrix}$ kg	SG 194 Mass and Inertia Matrix Calculation Document
$\mathbf{J}_f$	Added inertia matrix	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 10.062 & 0 \\ 0 & 0 & 10.062 \end{bmatrix}$ kg·m ²	SG 194 Mass and Inertia Matrix Calculation Document
$\mathbf{J}_s$	Stationary inertia matrix	$\begin{bmatrix} 0.1009 & -0.0604 & 0.1086 \\ -0.0604 & 8.1926 & -0.0015 \\ 0.1086 & -0.0015 & 8.1437 \end{bmatrix}$ kg·m ²	SG 194 Mass and Inertia Matrix Calculation Document
m_s	Stationary glider mass	58.852 kg	SG194 WA COAST-230ct-2013 data sheet
ρ	Density	1.023 $\frac{g}{cm^3}$	log files
P	Pressure	0.03 - 85.67 decibars	log files
T	Water temperature	9°C - 11°C	log files
T_0	Ambient temperature of water surface	10°C	log files

ii Variables

Variable	Description	Value / Range
t	Time stamp at which actuation states are commanded	0-20 s
pitchctl	Battery position along b_1 axis (aft positive)	cm
ϕ_{bat}	Battery rotation in yz -plane	rad or deg
VBD_{ctl}	Volume of oil in VBD	AD counts
$\mathbf{X}_0$ Initial state vector		
$\dot{\phi}_0$	Roll rate	deg/s (depends on Test ID)
$\dot{\theta}_0$	Pitch rate	deg/s (depends on Test ID)
$\dot{\psi}_0$	Yaw rate	deg/s (depends on Test ID)
ϕ_0	Roll angle	deg (depends on Test ID)
θ_0	Pitch angle	deg (depends on Test ID)
ψ_0	Yaw angle	deg (depends on Test ID)
u_0	Horizontal velocity (glide path frame)	0 m/s
v_0	Lateral velocity (glide path frame)	0 m/s
w_0	Vertical velocity (glide path frame)	m/s (depends on Test ID)
x_0	X distance in glide path frame	0 m
y_0	Y distance in glide path frame	0 m
z_0	Z distance / depth in glide path frame	m (depends on Test ID)

iii Measurements

Measurement	Description	Values / Range
$\dot{\phi}$	Roll rate	deg/s
$\dot{\theta}$	Pitch rate	deg/s
$\dot{\psi}$	Heading rate	deg/s
ϕ	Roll angle	deg
θ	Pitch angle	deg
ψ	Magnetic heading angle	deg
w	Vertical velocity (glide path frame)	m/s
z	Z distance / depth in glide path frame	m

iv Test Needs Matrix

Note: $\mathbf{X}_0 = [\dot{\phi} \ \dot{\theta} \ \dot{\psi} \ \phi \ \theta \ \psi \ u \ v \ w \ x \ y \ z]^T$

$$\mathbf{U} = [x_{bat} \ \phi_{bat} \ VBD_{ctl,AD}]^T$$

Test ID	Description	Conducted?	Justification
TDN -1	Zero initial velocity, neutrally buoyant; $\mathbf{X}_0 =$ $[\dot{\phi} \ \dot{\theta} \ \dot{\psi} \ \phi \ \theta \ \psi \ u \ v \ w \ x \ y \ z]^T =$ $[0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^T,$ $\mathbf{U} = [x_{bat} \ \phi_{bat} \ VBD_{ctl,AD}]^T =$ $[0 \ 0 \ 0]^T$, and $\Delta t = 30s$	Y	Simulation should not move without inputs that cause net forces or torques

TDN-2	Initial angular velocities, no hydrodynamic forces, repeated for each Euler angle; $\mathbf{X}_0 = [\dot{\phi} \ \dot{\theta} \ \dot{\psi} \ \phi \ \theta \ \psi \ u \ v \ w \ x \ y \ z]^T = [10 \ 0 \ 0 \ 10 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^T$, $\mathbf{U} = [x_{bat} \ \phi_{bat} \ VBD_{ctl,AD}]^T = [0 \ 0 \ 0]^T$, and $\Delta t = 30s$	Y	Simulation should not have any change in angular velocity with
TDN-3	Zero initial velocity, negatively buoyant, no hydrodynamic forces; $\mathbf{X}_0 = [\dot{\phi} \ \dot{\theta} \ \dot{\psi} \ \phi \ \theta \ \psi \ u \ v \ w \ x \ y \ z]^T = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^T$, $\mathbf{U} = [x_{bat} \ \phi_{bat} \ VBD_{ctl,AD}]^T = [0 \ 0 \ 0]^T$, and $\Delta t = 30s$	Y	Simulation should model gravity correctly
TDN-4	20 seconds snapshot of the dive phase of a past mission; $\mathbf{X}_0 = [\dot{\phi} \ \dot{\theta} \ \dot{\psi} \ \phi \ \theta \ \psi \ u \ v \ w \ x \ y \ z]^T$ where $x = y = 0$ since they are not compared, $z, \dot{z}, \theta, \phi, \psi, \dot{\theta}, \dot{\phi}, \dot{\psi}$ is from Mission data file; $\mathbf{U}$ components are from the Mission data file; Δt varies depending on first time step from the Mission data file		Full known-input known-output test of forces and dynamics subsystem
TDN-5	20 seconds snapshot of the climb phase of a past mission, same method as TDN-4		Full known-input known-output test of forces and dynamics subsystem for a different orientation than FD-4

v Test Matrix

Test ID	Run #	t_0	Δt	pitchctl	ϕ_{bat}	VBD_{ctl}	X_0	z	$\dot{z}$	θ	ϕ	ψ	$\dot{\theta}$	$\dot{\phi}$	$\dot{\psi}$	Filename/Notes